

2. Data Representation and Number System

Syeda Jannatul Naim

Lecturer | Dept. of CSE

World University of Bangladesh (WUB)

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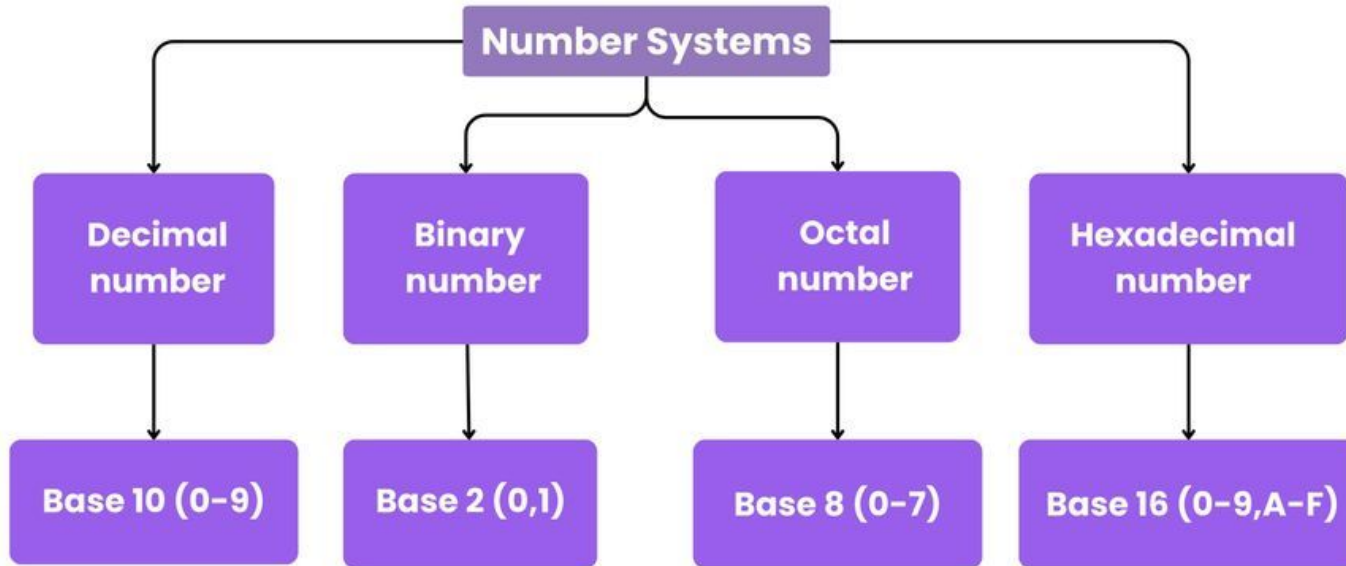
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1. Number System

Computers store and process all data in binary (0 and 1). Different number systems are used to make representation easier for humans.

1. Decimal Number System
 2. Binary Number System
 3. Octal Number System
 4. Hexadecimal Number System
- While humans use the decimal (base-10) system, computers internally use the binary (base-2) system.
 - Octal (base-8) and hexadecimal (base-16) are convenient shorthand notations for binary, making it easier for humans to read and work with machine-level data.

1. Number System



1. Number System

System	Base	Digits Used	Example	Primary Use Case
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	249	Human counting, mathematics.
Binary	2	0, 1	11111001	Internal computer operations (logic, memory).
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	371	Historically used in computing (Unix permissions).
Hexadecimal	16	0-9, A(10), B(11), C(12), D(13), E(14), F(15)	F9	Modern shorthand for binary (memory addresses, color codes).

Decimal System	Binary System	Octal System	Hexadecimal System
0	00000000	000	0
1	00000001	001	1
2	00000010	002	2
3	00000011	003	3
4	00000100	004	4
5	00000101	005	5
6	00000110	006	6
7	00000111	007	7
8	00001000	010	8
9	00001001	011	9
10	00001010	012	A
11	00001011	013	B
12	00001100	014	C
13	00001101	015	D
14	00001110	016	E
15	00001111	017	F

1. Number System

Key Concepts:

- **Base/Radix:** The number of unique digits (including zero) used in a number system. Determines the place value of each digit.
- **Bit:** Binary digit (0 or 1), the smallest unit of data.
- **Byte:** A group of 8 bits.
- **Nibble:** A group of 4 bits (half a byte).
- **MSB (Most Significant Bit/Byte):** The leftmost bit/byte, holding the greatest value.
- **LSB (Least Significant Bit/Byte):** The rightmost bit/byte, holding the smallest value

2. Utilization of Number System

2.1. Binary-based units in Computer

Unit	Symbol	Value
Kilobyte	KB	1 KB = 1024 bytes
Megabyte	MB	1 MB = 1024 KB
Gigabyte	GB	1 GB = 1024 MB
Terabyte	TB	1 TB = 1024 GB
Petabyte	PB	1 PB = 1024 TB
Exabyte	EB	1 EB = 1024 PB
Zettabyte	ZB	1 ZB = 1024 EB
Yottabyte	YB	1 YB = 1024 ZB

2. Utilization of Number System

2.2. Decimal-based units in Computer (storage)

Unit	Symbol	Value
Kilobyte	KB	1 KB = 1000 bytes
Megabyte	MB	1 MB = 1000 KB
Gigabyte	GB	1 GB = 1000 MB
Terabyte	TB	1 TB = 1000 GB

3. Character Encoding Systems

A system that maps characters (letters, digits, symbols) to numerical values that computers can store and process.

Key Concepts:

- **Code Point:** A numerical value assigned to a character
- **Encoding:** Process of converting characters to bytes
- **Decoding:** Process of converting bytes back to characters
- **Character Set:** Collection of characters the encoding supports

3. Character Encoding Systems

Different types character encoding system:

1. **ASCII (American Standard Code for Information Interchange) (7-bit)**
2. **Extended ASCII (8-bit)**
3. **BCD (Binary Coded Decimal)**
4. **EBCDIC (Extended Binary Coded Decimal Interchange Code)**
5. **Unicode -> Unicode Transformation Formats (UTF)**

Unicode Transformation Formats (UTF) are encoding schemes such as UTF-8, UTF-16, and UTF-32 used to store **Unicode** characters in binary form.

3. Character Encoding Systems

Feature	ASCII	BCD	EBCDIC	Unicode	UTF
Full Form	American Standard Code for Information Interchange	Binary Coded Decimal	Extended Binary Coded Decimal Interchange Code	Universal Character Encoding Standard	Unicode Transformation Format
Purpose	Character encoding	Numeric encoding	Character encoding	Universal character set	Encoding method for Unicode
Type	Character code	Numeric code	Character code	Character set	Encoding scheme
Bits Used	7-bit (8-bit extended)	4 bits per decimal digit	8-bit	21-bit code space	Variable (8/16/32 bits)
Data Represented	Letters, digits, symbols	Decimal digits (0–9)	Letters, digits, symbols	Characters from all languages	Unicode characters

3. Character Encoding Systems

Feature	ASCII	BCD	EBCDIC	Unicode	UTF
Language Support	English only	Numbers only	Limited languages	All languages	All languages
Storage Efficiency	Efficient	Inefficient	Moderate	Conceptual	Depends on UTF type
Compatibility	Widely compatible	Hardware/finance use	IBM systems	Platform independent	Platform independent
Modern Usage	Limited	Special purpose	Rare	Very common	Extremely common

END
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Chapter 2